

# Open Source & the Future of Innovation

**How Communities Shape Tomorrow's Technology**



# What if the smartest person in the room... isn't in the room?

Innovation used to look like:

**ONE COMPANY → ONE TEAM → ONE PRODUCT**

Open source changed the model:

**ONE PROBLEM → MANY MINDS → SHARED BUILDING**

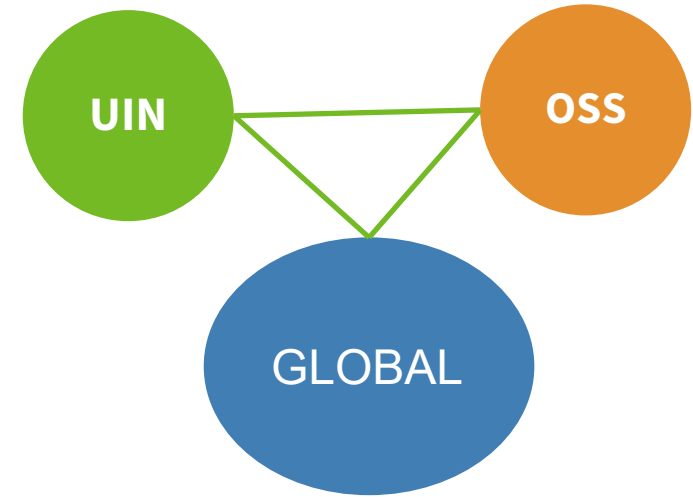
# A surprising fact about this campus

Open source is not new to UIN Sunan Kalijaga.

## 2016

**UIN Sunan Kalijaga hosted  
the openSUSE.Asia Summit.**

A global open-source community came to  
Yogyakarta to learn, share, and collaborate.



**The community model  
was already here.**

UIN's official Informatics page also documents KSL  
KUSUKA, focused on Linux & Open Source.

# The Innovation Shift

FROM CLOSED TEAMS TO OPEN COMMUNITIES



# The old model vs. the open model

## CLOSED / CENTRALIZED

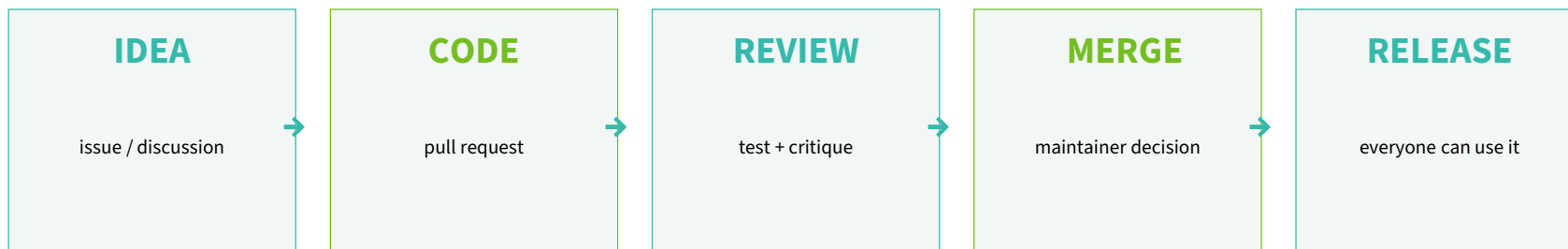
- Ideas stay inside one organization
- Small contributor pool
- Release → wait → next release
- Users mostly consume

## OPEN / COLLABORATIVE

- Ideas cross boundaries
- Global contributor pool
- Continuous review + iteration
- Users can become contributors

**The shift is not “free vs paid.” It is “centralized vs collaborative.”**

# How does a community actually build software?



**It is not “everyone coding at once.” It is a system of coordination.**

# Case study: Linux

A personal project that became global infrastructure.

**1991**

Linux began as a personal project by  
Linus Torvalds.

**GLOBAL**

Contributors and organizations now  
develop the kernel together.

**INFRASTRUCTURE**

Linux underpins servers, cloud,  
supercomputing and more.

**Lesson: a project can outgrow its creator when the community can improve it.**

# Case study: Python

The language is only the beginning. The ecosystem is the multiplier.

## READABLE

Lower barrier to entry

## PACKAGES

Reusable building blocks

## COMMUNITY

Libraries + docs + discussion

## APPLICATIONS

Data • science • AI • education

**Network effect: more users → more libraries → more users**

# The network is already enormous

Open source is no longer a niche developer practice.

**180M+**

developers on GitHub

**1.12B**

public / open-source  
contributions in 2025

**4.3M+**

AI-related repositories

**The important shift: innovation is becoming more networked.**

# Open Source + AI

THE NEXT INFLECTION POINT

# AI makes “open” more complicated

A model is more than a downloadable file.

**PARAMETERS**      Weights / learned values

**CODE**                      Training + inference + processing

**DATA**                      Training sources + processing

**The key question: What exactly is open?**

**OPEN  
WEIGHTS  
≠  
OPEN-SOURCE  
AI**

“Open” depends on what  
people can actually use,  
study, modify and share.

# Why would companies participate?

Open source can be a technical strategy, a business strategy, or both.

## SPEED

Reuse instead of rebuilding

## TALENT

Developers know the ecosystem

## ECOSYSTEM

Partners build around standards

## TRUST

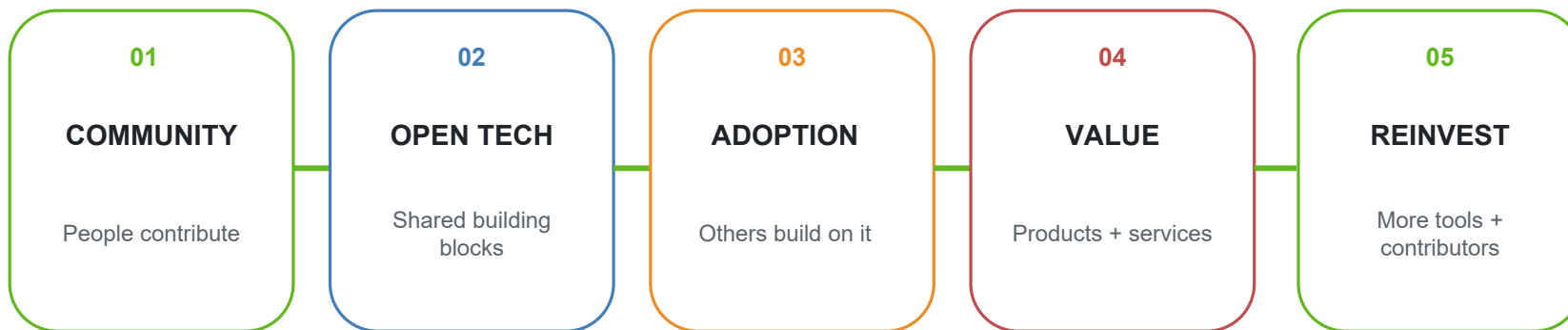
More eyes can inspect code

**Companies can compete on products while collaborating on foundations.**

# The hidden economics of Open Source



The code may be free. The ecosystem is not.



**Open source changes where value is created — and where value is captured.**

# But there is a human cost

Software may be open. Maintenance is not free.

## BUGS

Reports and fixes never stop.

## SECURITY

Critical projects need sustained attention.

## DOCS

Support and onboarding take time.

## GOVERNANCE

Someone decides what gets merged.

**The paradox: everyone can depend on it, while few people may maintain it.**

# Open does not automatically mean safe

Openness changes the risk profile; it does not remove risk.

## SECURITY

More eyes can find bugs.  
Vulnerabilities can also spread quickly.

## QUALITY

Community review is powerful.  
Not every contribution is correct.

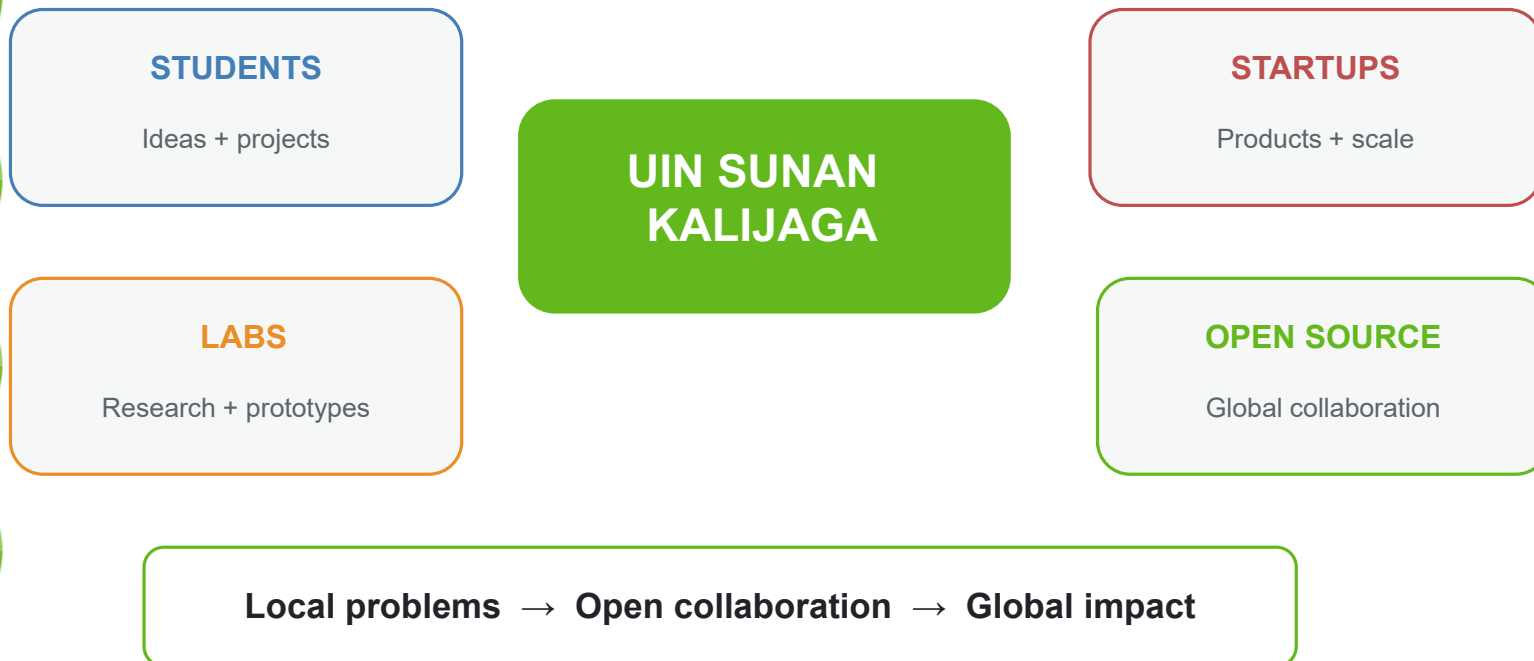
## GOVERNANCE

Projects still need rules:  
who reviews, merges and releases?

**Transparency helps. Practices, review and governance make it dependable.**

# What if UIN became an innovation node?

Not just a place where technology is learned — a place where technology is created.



# Imagine the year 2036...

What if the next breakthrough is built by people who never meet?



**No single company owns the whole journey. The innovation engine is the network.**

# Three shifts reshaping innovation

Questions to watch — not predictions.

01

**FROM PRODUCTS TO PLATFORMS**

Build once; let many others extend it.

02

**FROM TEAMS TO NETWORKS**

Expertise can move across universities, companies and countries.

03

**FROM CODE TO ECOSYSTEMS**

Models + tools + data + communities become the innovation unit.

The question is: where is this already happening?

# Who gets to build that future?

You do not need to start with a huge project.

01

## BUILD IN PUBLIC

A small repository can become a portfolio.

02

## CONTRIBUTE

Docs, testing, issues and translations count.

03

## CONNECT

Join a community beyond your classroom.

04

## SOLVE LOCAL PROBLEMS

Start with something people around you actually need.

**Your first contribution does not have to be code.**

# Thank You

Don't just download the future.

Help build it.

One final question:

What will you contribute?



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